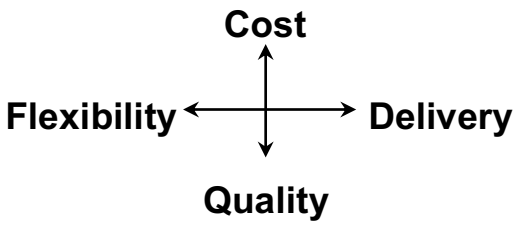
ISyE 315 DISCUSSION 1 (Chapters 1-2)

Definitions/Concepts	
Operations Management	The design, operation, and improvement of the systems that create and deliver the firm's primary products and services. "…creating great value to the customer while reducing the cost of delivering the good or service."
Transformations	Using resources to convert inputs into some desired outputs 6 Transformations: Physical, locational, exchange, storage, physiological, informational
Sustainability	The ability to meet current resource needs without compromising the ability of future generations to meet their needs *Triple Bottom Line*: Economic Prosperity, Environmental Stewardship, Social Responsibility
Core Capabilities / Competitive Dimensions	Skills that differentiate a manufacturing or service firm from its competitors • *Cost*: make it cheap • *Product Quality/Reliability*: make it good • *Delivery Speed (lead time):* make it fast • *Delivery Reliability*: deliver what/when promised • *Cope with changes in demand*: change volume • *Flexibility/new product introduction speed*: change it • *Product specific criteria*: support it
Trade-offs	 Understand the desire for balance: efficiency, effectiveness, value
Order Qualifiers	Set of dimensions used to screen a product or service as a candidate for purchase
Order Winners	Set of dimensions that differentiate the product or services of one firm from those of another. *Order winners can become order qualifiers over time
Risk Mitigation	Identify potential sources → Assess risks → Develop mitigation/correction plan Severity x Occurrence x Detection
KPI	Key Performance Indicators- need to measure in order to know what/where to improve "*Balanced Scorecard*" is desirable Ex: productivity, utilization, lead time, etc.
Productivity	A *relative* measure of how well resources are used (can also be partial or multifactor) Ex: Labor productivity can be looked at as net income or revenue per employee.

Equations	
Productivity	$Productivity = \frac{Output}{Input} = \frac{Goods\ and\ Service\ Produced}{Resources\ Used}$
Asset Productivity	$Receivable\ turnover\ = \frac{Annual\ Credit\ Sales}{Average\ Accounts\ Receivables}$ $Inventory\ turnover\ ratio = \frac{Cost\ of\ Goods\ Sold}{Average\ Inventory\ Value}$ $Asset\ turnover\ ratio = \frac{Revenue\ (or\ Sales)}{Total\ Assets}$
Input	$Labor\ cost\ +\ Material\ cost\ +\ Overhead\ costs$
Output	$Made\ Volume\ *\ Cost/unit$
Percent Change	$Percent\ Change = \frac{New - Old}{Old} * 100$

Design of Products and Services

Definitions, Concepts, and Equations	
Product Design Development	Planning → Concept Development → System Level Design → →Design Detail → Test/Refine → Ramp-up ***understand benefits of concurrent engineering - don't want "over the wall" operations***
Product Design Criteria	• Quality Function Development (often use house of quality) ○ QFD = a process that helps a company determine the product characteristics important to the consumer and to evaluate its own product in relation to others ○ *House of quality* = a matrix that helps a product design team translate customer requirements into operating and engineering goals ▪ blank row? = customer needs/requirements not met by the suggested engineering characteristics. May want or need to add appropriate considerations/features for customer satisfaction ▪ blank column? = the engineering characteristic doesn't affect the customers' needs. May be overdesigned and can be simplified. • VA/VE (value analysis/value engineering)

	○ Analysis with the purpose of simplifying products and processes by achieving equivalent or better performance at a lower cost • DFMA (Design or Manufacturing and Assembly) ○ Oriented toward the engineering of the product with an emphasis on reducing production costs (can you simplify or streamline?)
Breakeven Analysis	**P*N = F + V*N or revenue = expenses → N= F / (P-V)** F = fixed costs V = variable cost/unit N = number of units (volume) P = unit sale price *make vs. buy decision in semester project! Two main cases: • Find breakeven quantity for a single procedure/process • Find breakeven quantity when comparing two competitors. ○ Compare expenses with $F_1 + V_1N = F_2 + V_2N$